

Azure Cloud Interview Questions and Answers

1. What is Microsoft Azure?

Azure is Microsoft's cloud computing platform providing a range of services including compute, storage, and networking.

2. What are Azure Resource Groups?

A Resource Group is a container that holds related resources for an Azure solution.

3. What is Azure Virtual Machine?

A virtualized server running in Azure's cloud environment.

4. What are Azure Availability Zones?

Physically separate locations within a region to ensure high availability.

5. What is Azure App Service?

A PaaS offering to host web apps, REST APIs, and mobile backends.

6. What is Azure Blob Storage?

A service to store large amounts of unstructured data like text or binary.

7. What is Azure SQL Database?

A fully managed relational database service based on SQL Server.

8. What is Azure Active Directory (Azure AD)?

A cloud-based identity and access management service.

9. What is Azure Functions?

A serverless compute service that lets you run code on-demand.

10. What is a Virtual Network (VNet)?

A logically isolated network in Azure.

11. Difference between Azure Blob Storage and Azure File Storage?

Blob is for large unstructured data; File storage offers managed file shares.

12. What are ARM templates?

JSON files that define infrastructure and configurations for deployment.

14. Explain Azure Policy.

A service to create, assign, and manage policies for resource governance.

15. What is Azure Key Vault?

A service to securely store keys, passwords, certificates, and secrets.

16. What is Azure Kubernetes Service (AKS)?

A managed Kubernetes container orchestration service.

17. What is Azure Load Balancer?

A service that distributes network traffic to ensure high availability.

19. Explain Azure DevOps.

A set of tools for DevOps practices including CI/CD, boards, and repos.

21. How would you design a highly available architecture in Azure?

Use Availability Zones, Load Balancers and VM Scale Sets

22. What is the difference between Azure Scale Sets and Availability Sets?

Scale Sets automatically scale VMs; Availability Sets protect from hardware failures.

24. Explain Azure Traffic Manager.

A DNS-based traffic load balancer.

25. What is Azure API Management?

A service to create, publish, secure, and monitor APIs.

29. What is Azure Bastion?

A service for secure and seamless RDP/SSH access to VMs directly in Azure portal.

32. What are Azure Tags?

Key-value pairs to organize resources.

33. What is Azure Cost Management?

A tool to monitor, allocate, and optimize Azure spending.

34. What is Azure DevOps?

The process includes testing automation, continuous integration, as well as continuous delivery.

People with both development and operations skills work together and implement various tools for CI-CD and monitoring for quick response to customer's requirements and fix issues and bugs.

Azure DevOps is the new name for Microsoft VSTS (Visual Studio Team Services) and an application lifecycle management tool. It helps in project planning through Agile tools and templates. Other functionalities include version control of source code, management and execution of test plans, and management of branches.

Additionally, Azure DevOps also helps in solution deployments across various platforms with the help of Azure Pipelines and allows continuous integration and continuous deployment.

35. Why use DevOps?

Traditional software development always had a slow code deployment time after completion of development. And oftentimes, the Development Team and Operations Team or deployment team would get into arguments regarding its status blaming the server or the code for the issues. This is where DevOps enters with a solution.

DevOps facilitates the delivery of smaller features to clients in a quick and efficient manner and allows seamless software delivery.

36. How does DevOps work?

DevOps is the process of Operations and Development Engineers who work in collaboration in a complete project lifecycle, from the design and development to product releases and support.

37. What are the benefits of DevOps?

The major benefits of DevOps are:

- Customer satisfaction
- More engagement and collaboration between Development and Operation teams
- Faster code deployment in the market through continuous integration and delivery
- Faster operational support
- Higher efficiency
- Strong infrastructure and IT performance
- Continuous improvements and reduced failures
- Transparency between teams
- Constant monitoring and better adaption

38. What are Azure Boards?

Azure Boards is an Azure DevOps service. It helps manage the work in software projects and provides a wide range of functionalities like native support for Kanban and Scrum, customizable dashboards, and integrated reporting. Azure Boards include features like boards, sprints, work items, dashboards, backlogs, queries, etc.

39. What is Azure Repos?

This is a basic Azure DevOps interview question but it can be difficult to answer. Azure Repos is a version control system that can manage the different versions of a code and the code itself throughout the lifecycle of development. It is easy to track any changes in the code made by different teams. It also keeps a detailed record of these changes and the history for better coordination within the team. The changes are then merged at a later stage.

40. What are containers?

Containers provide the means to package software code, its configurations, dependencies, and packages into a single unit or object. Multiple containers can run on the same machine and share OS with other containers for running fast, reliable, and consistent deployments anywhere.

41. What are Azure Pipelines?

Azure DevOps Pipeline automatically develops and tests code projects. It is a service on the Azure cloud that works well with most project types and languages. This service helps in improving the availability of code projects to other users.

42. What are Azure DevOps Projects?

Azure DevOps Project is a simplified way to effectively bring existing code and Git repository for the creation of CI and CD pipeline to Azure.

43. What are the reasons to use CI and CD and Azure pipelines?

Make sure to go through this question as it is one of the important interview questions. Implementing CI and CD pipelines ensure quality and reliable code.

Azure pipelines are there to ensure a secure, easy, and quick way to automate project development processes and availability. They are entirely free for use in public projects and cost-effective for private ones (free 30 hours per month).

Below are a few reasons for using CI/CD and Azure pipelines:

- Supports any platform or language

- Enables work with open-source project>
- Building on Windows, Mac, and Linux machines
- Enables simultaneous deployment to various types of a target
- Integration with GitHub and Azure deployments

44. Which feature can be used for the development of a multi-tier application using Azure App Service web apps as the front end and Azure SQL database as the back end?

The appropriate option, in this case, is Application Map in Azure Application Insights. This is because it helps in identifying the performance bottlenecks as well as the failure hotspots in different components of the multi-tier applications.

Application components and the related dependencies are represented by the nodes on the map. Furthermore, it is also capable of providing the status for health KPI and alerts.

45. What is meant by the pull requests in Azure DevOps Repos?

Pull requests are used for reviewing as well as merging codes to a Git project. Pull requests enable the team to review the code and also provide feedback on it.

46. Tell us something about fault domains.

They are the logical hardware groups that can share the same power source as well as network. They are used for racking the data centers on-premise. These fault domains get automatically distributed by the Azure platform when VMs are designed using available sets. This approach limits network outages, physical hardware failures, etc.

47. Tell us something about the Update domains feature.

They are the logical hardware groups that can be rebooted as well as maintained at the same time. These updated domains get automatically distributed when the VMs are designed using available sets. This ensures our application instance, as well as the running of the Azure platform, is under periodic maintenance. Only a single order of updates can be rebooted at a time.

48. What is VNet?

The rendering of our cloud network refers to the VNet. Our launched cloud instances get isolated from the remaining resources with the help of VNet.

49. Elaborate self-hosted agent in Azure pipeline.

Self-hosted agents allow us to set up as well as management jobs on our own. We opt for self-hosted agents when we want more access to installing dependent software as per our needs and deployments. Also, the speed can be boosted by the persistence of machine-level caches and configuration from run to run.

50. Elaborate Microsoft-hosted agents in the Azure pipeline.

It is the by default Microsoft agents that get the jobs executed. From assigning the resources to installing and maintaining, everything gets done when we opt for this Microsoft-hosted agent.

51. Explain the use of an active directory in Azure.

It is used for access management. It helps in enabling access to the employees to particular products and services of our networks. Some major examples include Salesforce.com and Twitter. Application-built support in galleries is provided by this active directory.

52. Elaborate on the job role of Azure DevOps engineer.

An Azure DevOps engineer is responsible for designing and implementing a scalable, strong, realistic, and practical cloud solution that would work for the clients.

53. Explain the different phases of the DevOps lifecycle.

Azure DevOps interview questions cover every phase of the DevOps lifecycle separately or in a single question. The different phases in the DevOps methodology are:

Plan – The team discusses all the project requirements, like time for each stage, cost, and more.

Code – The software engineers write the small codes, known as units, according to the client's requirements.

Build – Involves the building of the units.

Test – The team tests if there are any mistakes. If yes, it is returned for a rebuild.

Integrate – Involves the integration of all the units of the codes.

Deploy – codeDevOpsNow is deployed on the client's environment.

Operate – Operations are performed on the code.

Monitor – This is the last phase wherein applications are monitored in the client's environment.

54. What is NSG?

NSG stands for Network Security Group that has a list of ACL (Access Control List) rules which either allows/denies network traffic to subnets or NICs (Network Interface Card) connected to a subnet or both.

55. Define Azure virtual machine scale sets

These are the Azure computation resources that can be used to deploy and manage sets of identical Virtual Machines (VMs).

These scale sets are configured in the same manner and are designed to support the autoscaling of the applications without the need for pre-provisioning of the VMs.

They help to build large-scale applications targeting big data and containerized workloads in an easier manner.

56. What do you understand about the “Availability Set”?

Availability Set is nothing but a logical grouping of VMs (Virtual Machines) that allows Azure cloud to understand how the application was developed for providing availability and redundancy.

Each VM in the availability set is assigned 2 kinds of domains by Azure:

Fault Domain: These define the grouping of VMs that would share a common power source and common network switch. The VMs within availability sets are separated across up to 3 fault domains by default. This separation of VMs in fault domains helps our applications to be available by reducing impacts of network outages, power interruptions, and certain hardware failures.

Update Domain: These indicate the grouping of VMs and underlying hardware which are eligible to be rebooted at the same time. Only one update domain can be rebooted at a time, however, the order of reboot does not proceed in a sequential manner. Before the maintenance of another update domain, the previously rebooted domain is given a recovery time of 30 minutes to ensure that the domain is up.

57. What is Azure Blob Storage?

Azure Blob storage is the object storage solution provided by Microsoft for the cloud. Blob stands for “Binary Large Object”. Blob-based storage is used to store massive unstructured data in terms of text or binary format. It is ideal for serving documents/images/audio/video/text directly to browser.

The data stored in the blob storage is accessible from anywhere in the world. The blobs are tied to user accounts by grouping them into containers. The Azure Blob Service has 3 components:

Storage Account: This can be a General Storage Account or Blob Storage Account registered in Microsoft Azure.

Container: Container is used for grouping blobs. We can store an unlimited number of blobs in a container. The name of the container should start in lowercase.

Blob: A blob is a Binary Large Object like a file or document of any type and size. There are 3 kinds of Blobs supported by Azure:

Block blobs: These are intended for text and binary files and can support up to 195GB, i.e up to 50k blocks of up to 4MB each.

Append blobs: These are used for appending operations like logging data in log files.

Page blobs: These are meant for frequent read/write operations.

24. What are IaaS, PaaS and SaaS?

IaaS: This stands for “Infrastructure as a Service” which provides a set of capabilities like OS, network connectivities, etc which are at the infrastructural level and are delivered as pay per use policy. The infrastructure is used for hosting applications. Examples include Azure VM, VNET, etc.

PaaS: PaaS stands for “Platform as a Service” which is mostly about underlying infrastructure abstraction to the developers for enabling quicker development of the applications without the need for worry about hosting management. Examples include Azure web apps, Storage services, cloud services, etc.

SaaS: SaaS stands for “Software as a Service” and are those applications which are delivered using the service delivery model where the applications are simply consumed and used by an organization. These applications are generally mobilized by making the organization pay for their usage or through ads. Examples include applications like Office 365, Gmail, SharePoint Online, and so on.

58. Define Azure Functions in detail.

The Azure Functions is a serverless code computation service that allows you to run code without a server on demand, such as Events and External-Invoke. They are stateless and short-lived, and Azure Functions may automatically scale up in response to the request. They tend to speed up the development process by avoiding the need to perform any integration coding for you to connect to other services. They also offer Azure Application Insights for monitoring and evaluating code performance, which aids in the identification of bottlenecks and failure locations throughout the application's components.

59. Explain Service Bus Queue and Storage Queue.

Azure Service Bus Queues belong to the Azure messaging framework and include queuing, publishing, subscribing, etc. They are part of the Service Bus and can pass messages through to other Queues and Topics. The Azure Service Bus Queues feature a built-in dead-letter queue and allow you to choose a timeline for messages so they can last as long as you want them to! They connect applications or parts of applications that cover different communication protocols, data treaties, trust domains, or security protocols.

Azure Storage Queues belong to the Azure storage framework and are easy to use. They allow easy debugging by using the local Azure Storage Emulator. The set of Azure Storage Queue tools enables you to take a quick look at the top 32 messages and visualize the contents of those belonging to XML/JSON right from Visual Studio. Another feature of storage queues that ensures smooth development and QA operations is that the contents of these queues can be emptied when needed. Authenticated HTTP or HTTPS calls allow you to access the queue messages regardless of your geographical location. Queue messages have a maximum capacity of 64 KB and can hold millions of messages depending upon the storage account's overall capacity limit.

60. Can you specify the storage limit associated with a virtual machine?

When it comes to the storage limit associated with virtual machines, each data disk has a maximum capacity of 1 TB. The amount of data disks one can use is determined by the virtual machine's size. Azure Managed Disks is a new and highly recommended disk storage option for Azure Virtual Machines for long-term data storage. Each Virtual Machine can have many Managed Disks. Premium and Standard Managed Disks are two types of long-term storage options offered by Managed Disks. Also, storage for the OS disk and any data disks can be provided via Azure storage accounts. Each disk is a page blob that is stored in a .vhd file format.

61. How does Azure Traffic Manager differ from Azure Load Balancer?

Azure Traffic Manager is mainly responsible for sending traffic globally on the basis of dynamic principles. This leads to an enhanced user experience that reflects how your application is distributed worldwide. Your public endpoints will also experience benefits such as high accessibility and responsiveness. For diverse application objectives and spontaneous recovery systems, Traffic Manager offers a variety of traffic-routing mechanisms and endpoint tracking solutions.

The Azure Load Balancer handles the routing of traffic within a certain region. It is used together with Azure Traffic Manager, which directs traffic to a region between virtual machines. When the two are coupled, you get global traffic control with local backup. The

Azure Load Balancer service offers a high-performance, low bandwidth Layer 4 load-balancing solution for all UDP and TCP protocols. It can ensure the smooth handling of millions of queries per second while maintaining high availability.

62. Define Azure SQL Database?

Azure SQL Database is a fully managed PaaS database server that keeps track of most database management tasks like data upgradations, patching, data backups, etc., without the need for human intervention. Azure SQL Database always runs on the most recent stable version of the Microsoft SQL Server database system, and its built-in PaaS capabilities allow you to concentrate on the domain-specific database management and performance activities that play a major role in upscaling your business. It also enables you to develop a highly accessible and rich in performance data storage layer for Azure apps and solutions.

63. Discuss the different types of backups available in Azure?

To maintain the high availability of your data as well as storage, Azure Backup supports three methods of backup-

- i. Locally redundant storage (LRS) copies your data three times in a centralized storage unit within the same region. LRS is a cost-effective option for safeguarding data against local hardware breakdowns.
- ii. Geo-redundant storage (GRS) is the standard and preferred backup mode that replicates your data to a secondary region far away from the primary location of the source data. GRS is more expensive than LRS, but it provides greater data resilience, even in the event of a local disruption.
- iii. Zone-redundant storage (ZRS) backs up data in availability zones, ensuring data retention and durability in the same zone. You can back up your essential activities that involve data retention and must run without disruption since ZRS has zero latency.

64. What do you mean by Azure Resource Manager Templates (ARM)?

Azure Resource Manager (ARM) templates are JSON files mainly used to implement infrastructure as code for your Azure solutions. The template outlines your project's infrastructure and configuration. A declarative syntax is used in the template, which allows you to declare what you want to deliver without the use of any actual programming/coding. The template must include the resources to launch as well as their attributes.

65. Briefly discuss the Azure Kubernetes Service (AKS).

Azure Kubernetes Service (AKS) aims at the easy installation of a hosted Kubernetes cluster on Azure. It is solely responsible for managing containerized applications. It's an open-source solution for scaling, automatic deployment, and managing traffic. Azure deals

with important functions like health diagnosis and management as a managed Kubernetes service. You solely manage and maintain the agent nodes because the Kubernetes masters are controlled by Azure. As a result, AKS is completely free; you only have to pay for the agent nodes in your clusters.

66. What is meant by autoscaling in Azure?

Autoscale is an inbuilt feature of Cloud Services, Virtual Machine Scale Sets, and Websites that enables applications to expand to adapt to changing demand. Scaling out refers to increasing the number of instances in a system. Scaling up in Windows Azure is also possible by using larger role instances rather than additional role instances. You may balance the performance of your Windows Azure application against its operational costs by adding and removing role instances while it is running. The amount of manual work necessary in dynamically scaling an application is reduced with an autoscaling solution.

67. Briefly define SQL Azure firewall.

SQL Azure Firewall is a cloud-based service that enables you to determine IP addresses and filter traffic to a VNet. It is designed for high availability and unlimited cloud scalability, as well as protecting inbound and outbound traffic to the VNet.

SQL Azure firewall is a Microsoft Azure Platform cloud computing database service that allows you to host and access SQL databases.

68. What is Build Pipeline?

The Source Code stored in Repo will be checked for any errors and if there is no error an artifact will be generated in the Drop Folder

69. What is Release Pipeline?

The artifact generated by build pipeline will be deployed to cloud or virtual machine by using the Release pipeline.

70. Explain variable and variable groups in Azure DevOps?

Variables allow you to store some data that can be used across pipelines. All variables are mutable and stored as strings.

Variable Groups provide the ability to use variables across multiple pipelines. You can store secrets in variable or variable groups.

71. Explain git rescan, stage, commit and push?

Rescan: It will check whether there any new code changes done by the programmers.

Stage: All the changes will be staged here.

Commit: All staged changes will be committed here and will be ready for the push into repos.

Push: All the committed changes will be pushed to Azure Repo.

72. What is Service Connection?

Service Connections are used in Azure DevOps Pipelines to connect to external services, like Azure, GitHub, Docker, Kubernetes, and many other services.

73. What is Key vault?

Azure Key Vault is a cloud service for securely storing and accessing secrets. API keys, passwords, certificates

74. What is Deployment Group?

A deployment group is a logical set of deployment target machines that have agents installed on each one. Deployment groups represent the physical environments; for example, "Dev", "Test", or "Production" environment. In effect, a deployment group is just another grouping of agents, much like an agent pool.

75. What are routing methods in traffic manager azure?

- Priority traffic-routing method
- Weighted traffic-routing method
- Performance traffic-routing method
- Geographic traffic-routing method
- Multi value traffic-routing method
- Subnet traffic-routing method

76. What is an application insight in azure?

Application Insights is an extension of Azure Monitor and provides Application Performance Monitoring (also known as "APM") features. APM tools are useful to monitor applications from development, through test, and into production in the following ways:

- Proactively understand how an application is performing.
- Reactively review application execution data to determine the cause of an incident.

77. What is dns in azure?

Azure DNS is a hosting service for DNS domains that provides name resolution by using Microsoft Azure infrastructure. By hosting your domains in Azure, you can manage your DNS records by using the same credentials, APIs, tools, and billing as your other Azure services.

78. What is SonarCloud?

SonarCloud is a cloud-based code analysis service designed to detect coding issues in 26 different programming languages.

79. What is service health?

Service Health provides a personalized view of the status of your Azure services and regions, as well as information about current incidents, planned maintenance and health advisories.

80. What are pods in kubernetes?

A pod is the smallest execution unit in Kubernetes. A pod encapsulates one or more applications. Pods are ephemeral by nature, if a pod (or the node it executes on) fails, Kubernetes can automatically create a new replica of that pod to continue operations.

81. What is Docker?

Docker is an open source platform that enables developers to build, deploy, run, update and manage containers—standardized, executable components that combine application source code with the operating system (OS) libraries and dependencies required to run that code in any environment.

82. what is a Container?

Containers simplify development and delivery of distributed applications. They have become increasingly popular as organizations shift to cloud-native development and hybrid multicloud environments.

83. What is Docker Image?

A Docker image is a file used to execute code in a Docker container. Docker images act as a set of instructions to build a Docker container, like a template. Docker images also act as the starting point when using Docker. An image is comparable to a snapshot in virtual machine (VM) environments.

84. What is container registry?

Azure Container Registry allows you to build, store, and manage container images and artifacts in a private registry for all types of container deployments. Use Azure container registries with your existing container development and deployment pipelines.

85. What is container instance?

Azure Container Instances is a solution for any scenario that can operate in isolated containers, without orchestration. Run event-driven applications, quickly deploy from your container development pipelines, and run data processing and build jobs.

86. What is kubectl used for?

The Kubernetes command-line tool, kubectl, allows you to run commands against Kubernetes clusters. You can use kubectl to deploy applications, inspect and manage cluster resources, and view logs.

87. What is azure cli?

The Azure Command-Line Interface (CLI) is a cross-platform command-line tool to connect to Azure and execute administrative commands on Azure resources. It allows the execution of commands through a terminal using interactive command-line prompts or a script.

88. What is DockerFile?

A Dockerfile is a text document that contains all the commands a user could call on the command line to assemble an image. This page describes the commands you can use in a Dockerfile

89. How do I write a YAML file for Kubernetes?

To create a Kubernetes pod with YAML, you first create an empty file, assign it the necessary access permissions, and then define the necessary key-value pairs. The important ones are the apiVersion, the kind (pod), name, and the containers within the pod.

90. What is service in Kubernetes?

A Kubernetes service is a logical abstraction for a deployed group of pods in a cluster (which all perform the same function). Since pods are ephemeral, a service enables a group of pods, which provide specific functions (web services, image processing, etc.) to be assigned a name and unique IP address (clusterIP).

91. What is Name Server?

A name server refers to the server component of the Domain Name System (DNS), one of the two principal namespaces of the Internet. The most important function of DNS servers is the translation (resolution) of human-memorable domain names (example.com) and hostnames into the corresponding numeric Internet Protocol (IP) addresses (192.0.2.1)

92. What is custom domain in azure app service?

Configuring Custom Domain. Adding a custom domain to Azure App Service requires you to have access to your domain provider service where you will need to update DNS records to confirm that you own the domain. You can choose to use a root domain or a sub domain to redirect to your application based on your requirement.

93. What is bastion azure?

Azure Bastion is a fully managed service that provides more secure and seamless Remote Desktop Protocol (RDP) and Secure Shell Protocol (SSH) access to virtual machines (VMs) without any exposure through public IP addresses.

94. Define Azure Service Level Agreement (SLA)?

- The Azure SLA is a contract that ensures or guarantees that when two or more role instances of a role are deployed on Azure, access to that cloud service is guaranteed for at least 99.95% of the time.
- It also states that if the role instance process is not in the running state, then the detection of such processes and corrective action for the same will be taken 99.9% percent of the time.
- If the mentioned guarantees are not satisfied at any point in time, then Azure credits a percentage of monthly fees to us depending on the pricing model of the respective Azure services.

95. What is VM Clone?

You can create a full copy or a linked copy of an existing VM. This copy is called a clone.

96. what is snapshot in azure?

A snapshot is a full, read-only copy of a virtual hard disk (VHD).

97. What is DNS Zone?

A DNS zone is used to host the DNS records for a particular domain. To start hosting your domain in Azure DNS, you need to create a DNS zone for that domain name. Each DNS record for your domain is then created inside this DNS zone.

98. what is sso in azure?

Azure SSO is a cloud-based SSO solution that helps organizations simplify the management of user identities and access control in the cloud.

99. What is APIM Service?

Azure API Management is a hybrid, multicloud management platform for APIs across all environments. As a platform-as-a-service, API Management supports the complete API lifecycle

100. What is Azure AD Connect?

Azure AD Connect (now referred to also as Azure AD Connect “Classic”) is a Microsoft brand that is used to migrate data from on-premises Active Directory to Azure Active Directory in a seamless way.

101. What is Terraform and why it is used?

Terraform is an IAC tool, used primarily by DevOps teams to automate various infrastructure tasks. The provisioning of cloud resources, for instance, is one of the main use cases of Terraform. It's a cloud-agnostic, open-source provisioning tool written in the Go language and created by HashiCorp.

102. What are basic Terraform Commands?

terraform init - Initialize Terraform

terraform validate - Validates the code

terraform plan - Review the configuration and verify that the resources that Terraform is going to create

terraform apply - creates the Terraform configuration in Cloud

103. What is Docker?

Answer: Docker is an open-source platform designed to automate the deployment, scaling, and management of applications using containerization. Containers package an application with its dependencies, ensuring consistency across multiple environments.

104. What is the difference between a Docker container and a virtual machine?

Answer:

- Docker Container: Shares the host OS kernel, lightweight, starts quickly, consumes less memory.
 - Virtual Machine: Has a full guest OS, heavier, slower to boot, consumes more resources.
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105. What is Dockerfile?

Answer: A Dockerfile is a text file containing instructions to build a Docker image. It defines the base image, environment variables, commands to run, ports to expose, and more.

106. What is the difference between Docker image and Docker container?

Answer:

- Docker Image: A static, read-only template used to create containers.
 - Docker Container: A running instance of an image with a writable layer.
-

107. What is Docker Hub?

Answer: Docker Hub is Docker's official cloud-based registry where you can find and share container images. It includes official images, user-contributed images, and automated builds.

108. How do you persist data in Docker?

Answer: You can persist data using:

- Volumes: Managed by Docker and stored outside the container filesystem.
 - Bind mounts: Mounts a specific path from the host machine.
-

109. What is the purpose of ENTRYPOINT and CMD in Dockerfile?

Answer:

- CMD sets default arguments for the container when no command is provided.
 - ENTRYPOINT sets the executable that always runs. You can combine them to override parameters but not the command.
-

110. What is a Docker volume?

Answer: A Docker volume is a persistent storage mechanism used to store data outside the container lifecycle. Volumes can be shared across containers and are stored in Docker's local storage.

111. What is a multi-stage build in Docker?

Answer: Multi-stage builds allow you to use multiple FROM statements in a Dockerfile to separate build and production environments, reducing final image size by excluding unnecessary dependencies.

112. How does Docker Compose work?

Answer: Docker Compose is a tool for defining and running multi-container Docker applications using a docker-compose.yml file. It lets you configure services, networks, and volumes declaratively.

113. How do you reduce Docker image size?

Answer:

- Use a smaller base image (like alpine).
- Use multi-stage builds.

- Clean up temporary files and package caches (`rm -rf /var/lib/apt/lists/*`).
 - Combine commands to reduce layers.
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114. How does Docker handle networking?

Answer: Docker provides several network drivers:

- bridge (default),
- host (shares host network)

115. How can you run Docker containers in Azure?

Answer:

Docker containers can run in Azure using several services:

- Azure Container Instances (ACI): Lightweight and serverless container execution.
 - Azure Kubernetes Service (AKS): For orchestrated and scalable container deployments.
 - Azure App Service: Supports Docker containers for web apps.
 - Azure Virtual Machines: Full Docker environment via VM installation.
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116. What is Azure Container Registry (ACR)?

Answer:

ACR is a private Docker registry service on Azure. It allows you to store and manage your own Docker container images securely and integrate with ACI, AKS, and CI/CD pipelines.

117. How do you deploy a Docker container to Azure Container Instances (ACI)?

Answer:

You can deploy using:

- Azure CLI:

bash

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```
az container create --name mycontainer --image myimage --resource-group mygroup  
--registry-login-server myregistry.azurecr.io
```

- Azure Portal

- ARM templates or Bicep
- Terraform or other IaC tools

118. What is the difference between ACR and Docker Hub?

Answer:

- ACR: Private, Azure-integrated, supports RBAC, geo-replication, and better performance in Azure-based workflows.
- Docker Hub: Public or private repositories, widely used, but not deeply integrated with Azure.

119. How does Azure Kubernetes Service (AKS) work with Docker?

Answer:

AKS uses container images (typically Docker or OCI format) to deploy applications at scale. You build Docker images, push them to ACR or Docker Hub, and configure AKS to pull and run them.

120. How can you automate Docker deployments in Azure using DevOps pipelines?

Answer:

You can use Azure Pipelines to:

- Build Docker images with a Dockerfile.
- Push to Azure Container Registry.
- Deploy to AKS, ACI, or App Service using YAML pipelines or classic release pipelines.

121. How do you deploy multiple containers together in Azure?

Answer:

- Use Docker Compose with Azure App Service (Web Apps for Containers) (via custom startup scripts).
- Use AKS with a Kubernetes manifest (YAML) to define multiple Pods/Deployments.
- Use Azure Container Apps (PaaS model) for simpler, microservice deployments with multiple containers.

122. What are ACR Tasks and how are they useful?

Answer:

ACR Tasks automate building, testing, and pushing Docker images to ACR when:

- The source code changes (via GitHub webhook).
- A base image is updated (e.g., a new Node.js base image).
- A schedule is triggered.

123. Can you use Docker containers in Azure App Service?

Answer:

Yes, Azure App Service supports custom containers:

- Hosted from Docker Hub, ACR, or any public/private registry.
- Supports multi-container apps using docker-compose.yml.

124. How do you implement CI/CD with Docker in Azure DevOps?

Answer:

- Use a pipeline with the following stages:
 1. Build the Docker image using the Docker@2 task.
 2. Push the image to ACR.
 3. Deploy to AKS or ACI using kubectl, helm, or AzureCLI@2 tasks.
- Use YAML or classic pipelines with service connections.

125. What is Kubernetes?

Kubernetes (K8s) is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications.

126. What are Pods in Kubernetes?

A Pod is the smallest deployable unit in Kubernetes. It can hold one or more containers that share storage, network, and a specification for how to run them.

127. What is the difference between a Pod and a Deployment?

- Pod: A single running instance of a container.
- Deployment: A controller that manages Pods, handles updates, scaling, and self-healing.

128. What is a Service in Kubernetes?

A Service is an abstraction that exposes a set of Pods as a network service. It enables communication between components within and outside the cluster.

129. What are the types of Services in Kubernetes?

- ClusterIP (default): Internal access only.

- NodePort: Exposes service on each Node's IP at a static port.
- LoadBalancer: Uses cloud provider's load balancer.

130. What is a ConfigMap and a Secret?

- ConfigMap: Stores non-sensitive configuration data.
- Secret: Stores sensitive information like passwords, tokens, and keys.

131. What is the role of etcd in Kubernetes?

etcd is a distributed key-value store that stores all cluster state data, including configuration, secrets, and node information.

132. What is a Namespace in Kubernetes?

A Namespace is a logical partition within a Kubernetes cluster, used to isolate resources between different teams or projects.

133. How does Terraform work with Azure?

Terraform uses the Azure Provider (azurerm) to provision resources in Azure. It interacts with Azure's REST APIs to create, update, or destroy resources defined in .tf files.

134. What are the basic components of a Terraform configuration?

Provider – Defines the cloud platform (e.g., azurerm).

Resources – Describe infrastructure objects (VMs, storage accounts, etc.).

Variables – Parameterize configurations.

Outputs – Return values after execution.

State File – Tracks resource deployments.

135. How do you authenticate Terraform with Azure?

Terraform supports multiple authentication methods:

Azure CLI (az login)

Service Principal with client secret or certificate

Managed Identity (for use in Azure DevOps or VMs)

136. What is the purpose of the Terraform state file (terraform.tfstate)?

It keeps track of the resources Terraform manages. This enables Terraform to detect changes and perform updates incrementally. The state file should be secured and can be stored remotely in Azure Storage for team environments.

137. What is the difference between terraform plan and terraform apply?

- terraform plan: Shows what Terraform will do without making changes.
- terraform apply: Applies the changes to your infrastructure.

138. How do you perform environment separation (e.g., dev, staging, prod) in Terraform?

- Use workspaces, variable files (*.tfvars), or directory structures.
- Separate state files per environment.

139. How do you manage secrets in Azure DevOps?

Answer:

- Use Azure Key Vault and link it to the pipeline.
- Use Pipeline Secrets/Variables with "Keep secret" enabled.
- Avoid hardcoding secrets in YAML files.

140. What is an Agent in Azure DevOps?

Answer:

An agent is a machine (hosted or self-hosted) that runs jobs in a pipeline.

- Microsoft-hosted: Managed by Microsoft.
- Self-hosted: You maintain it on your VM or infrastructure.

141. How can you deploy an app to Azure using Azure DevOps?

Answer:

- Define CI pipeline to build the app.
- Define CD pipeline to deploy it using tasks like Azure Web App Deploy or Azure CLI.
- How do you set up multi-stage pipelines in Azure DevOps?

Answer:

Use stages, jobs, and steps in a YAML pipeline to define multiple environments like build, test, and deploy:

- stages:
 - - stage: Build
 - - stage: Deploy

142. How do you implement infrastructure as code (IaC) in Azure DevOps?

Answer:

Use tools like Terraform, ARM templates, or AZ CLI integrated with pipelines to deploy infrastructure. Terraform tasks can be used directly in the pipeline.

143. What is Azure Application Insights?

Azure Application Insights is an Application Performance Management (APM) service that enables you to monitor your live applications. It helps detect anomalies, diagnose performance issues, and understand user behavior using telemetry data.

144. What kind of telemetry data does Application Insights collect?

- Requests
- Dependencies
- Exceptions
- Page views (for web apps)
- Custom events
- Traces (logs)
- Performance counters

145. How can you integrate Application Insights into an application?

You can integrate it by:

- Installing the Application Insights SDK
- Adding the Instrumentation Key (or connection string)
- For ASP.NET, ASP.NET Core, Node.js, Java, etc., SDKs or agents are available
- For some Azure services (like App Services), it can be enabled without code changes

146. What is an Instrumentation Key (ikey)?

The Instrumentation Key is a unique identifier used by Application Insights to associate telemetry data with your specific Application Insights resource.

147. Can Application Insights monitor non-Azure apps?

Yes. Application Insights can monitor on-premise, hybrid, and other-cloud hosted applications as long as the telemetry SDK is integrated.

148. What is sampling in Application Insights?

Sampling reduces the volume of telemetry data sent to Application Insights to lower costs and improve performance. Types include:

- Fixed-rate sampling
- Adaptive sampling – automatically adjusts the sampling rate to maintain a target number of events per second.

149. How do you track custom events or metrics?

You can use methods like:

```
TelemetryClient.TrackEvent("CustomEventName");
```

```
TelemetryClient.TrackMetric("MetricName", value);
```

This is useful for tracking business-specific operations or KPIs.

150. What is Live Metrics Stream?

Live Metrics Stream shows real-time performance data like incoming requests, failures, and exceptions. It's useful for monitoring live deployments or debugging production issues quickly.

151. What is Application Map in Application Insights?

Application Map provides a visual topology of your application architecture, showing components (e.g., microservices) and their dependencies, along with performance and failure information.

152. How do you create alerts in Application Insights?

You can create alerts based on:

- Metrics (e.g., server response time, failure count)
- Logs (via KQL queries)
Alerts can trigger actions like sending emails, calling webhooks, or triggering Azure Functions.

153. How can you identify performance bottlenecks in Application Insights?

You can identify performance issues by:

Navigating to the Performance blade in Application Insights.

Reviewing slowest requests, their durations, and response times.

Inspecting dependencies to see if third-party calls (e.g., SQL, REST APIs) are causing slowness.

Using End-to-End Transaction View to trace the request lifecycle.

154. What is the End-to-End Transaction Diagnostics view, and how is it used?

This view visualizes how a single request flows through your system. It includes:

Incoming request

Dependencies (e.g., DB calls, external services)

Exceptions (if any)

Time taken by each step

This helps identify where exactly latency or errors are occurring.

155. How do you investigate a spike in failed requests?

Steps to investigate:

Go to the Failures blade.

Filter by operation name, status code, or exception type.

Use Search or Analytics (KQL) to look at details.

Use Drill into operation to get end-to-end details.

Check related traces/logs for context.

156. What is the Failures blade in Application Insights?

The Failures blade provides a visual overview of:

Failed requests

Exception trends

Most frequent failure types

Failure breakdown by operation or dependency

Useful for identifying patterns and root causes of application errors.

157. How do you correlate telemetry across different components or services?

Application Insights uses operation ID and parent/child telemetry IDs to correlate events.

You can:

Use Search and click on a request to see related dependencies and exceptions.

Use Application Map to view dependencies between services and their health.

158. How do you investigate an exception using Application Insights?

Go to Failures → Exceptions.

Identify the exception type and frequency.

Click on individual exceptions to see stack traces, custom properties, and related telemetry.

Use "Drill into" to trace the root cause across services.

159. How can Live Metrics help during an investigation?

Live Metrics Stream provides real-time insights:

Requests per second

Failures

Server performance metrics

Custom telemetry (if configured)

It's useful during a live incident or deployment to detect immediate impact or degradation.

160. How can you write a query to investigate slow requests?

KQL example:

```
pgsql
```

```
Copy
```

```
Edit
```

```
requests
```

```
| where duration > 2s
```

```
| order by duration desc
```

```
| project timestamp, name, duration, success, url
```

This identifies the slowest requests, helping to prioritize investigation.

161. What is Application Map and how does it help in investigations?

Application Map shows a topological view of your application architecture. It highlights:

Components (App Services, APIs, DBs)

Health indicators (latency, failures)

Call relationships

Great for understanding upstream/downstream dependencies during incidents.

162. Can Application Insights help in root cause analysis (RCA)? How?

Yes. It provides:

End-to-end traces for requests

Stack traces for exceptions

Dependency timings

Related logs and metrics

Visual maps for system interaction

You can combine these insights to trace problems back to their origin.

163. What is Azure Traffic Manager?

Azure Traffic Manager is a DNS-based traffic load balancer that enables you to distribute traffic optimally across global Azure services, improving availability and responsiveness.

164. How does Azure Traffic Manager work?

It uses DNS to direct client requests to the most appropriate endpoint based on the routing method you choose, such as performance, priority, geographic, etc.

165. What are the routing methods supported by Azure Traffic Manager?

- Priority – Directs traffic to the primary endpoint; others are backup.
- Weighted – Distributes traffic based on assigned weights.
- Performance – Routes traffic to the lowest-latency endpoint.
- Geographic – Routes traffic based on geographic location of users.
- Multivalue – Returns multiple healthy endpoints.

166. What are endpoints in Traffic Manager?

Endpoints are the resources (Azure VMs, web apps, external services) that Traffic Manager directs traffic to. Types:

- Azure endpoints
- External endpoints (non-Azure)
- Nested endpoints (Traffic Manager profiles)

167. What happens if all endpoints are unhealthy?

Traffic Manager continues to return DNS responses, but since all endpoints are unhealthy, users may experience failures. You should plan for redundancy.

168. Can Traffic Manager route traffic to on-premises servers?

Yes. Use External endpoints to route traffic to on-premises or non-Azure services by specifying their public DNS names or IP addresses.

169. What are environment variables in Azure App Service?

Environment variables in Azure App Service are configuration settings used to pass data such as connection strings, secrets, or runtime-specific values to your application without hardcoding them.

170. How do you set environment variables in Azure App Service?

You can set environment variables through the Azure Portal:

- Go to your App Service > Configuration > Application settings.
- Add key-value pairs under Application settings or Connection strings.

171. Where can you access environment variables in your code?

- In .NET: `Environment.GetEnvironmentVariable("KEY")`
- In Node.js: `process.env.KEY`
- In Python: `os.environ.get("KEY")`
- In Java: `System.getenv("KEY")`

172. What is the difference between Application Settings and Connection Strings in Azure App Service?

- Application Settings: Used for general environment variables.
- Connection Strings: Special category for database connection strings. Azure automatically injects them into the environment with a prefix like `SQLCONNSTR_`, `MYSQLCONNSTR_`, etc.

173. Do changes to environment variables in App Service restart the app?

Yes. Any changes to application settings or connection strings trigger an automatic restart of the App Service.

174. Are Azure App Service environment variables secure?

- Application settings are encrypted at rest and decrypted at runtime.
- Connection strings are also encrypted and handled securely.
- However, they are visible in the Azure portal, so sensitive data (like secrets) should ideally be stored in Azure Key Vault and referenced securely.

175. How can you access Azure Key Vault secrets as environment variables in App Service?

- Add a Key Vault reference in Application Settings:
- Format: @Microsoft.KeyVault(SecretUri=<Secret-URI>)
- The app must be assigned Managed Identity and granted access to the Key Vault.

176. Can you override environment variables at runtime?

Yes, but:

- Changes made inside the app (e.g., via code) only affect the current process and are lost after a restart.
- To make persistent changes, use the Azure Portal or CLI.

177. What are some common environment variables automatically set by Azure App Service?

Examples:

- WEBSITE_SITE_NAME: Name of the App Service
- WEBSITE_INSTANCE_ID: Unique identifier for the instance
- HOME: Path to the home directory
- WEBSITE_CONTENTSHARE: Used for shared storage across instances

178. What is Azure SQL Database?

Azure SQL Database is a fully managed, cloud-based relational database service built on Microsoft SQL Server. It provides high availability, scalability, and security without the need to manage hardware or software.

179. How is Azure SQL Database different from SQL Server on-premises?

- Fully managed PaaS service.
- Built-in high availability and backups.
- No OS/SQL Server maintenance needed.
- Scales easily on demand.

180. What are the main components of Kubernetes architecture?

Master Node (Control Plane):

- API Server
- Scheduler
- Controller Manager
- etcd (Key-value store)

Worker Nodes:

- Kubelet
- Kube Proxy
- Container Runtime (e.g., Docker, containerd)

181. What is a Pod?

A Pod is the smallest deployable unit in Kubernetes. It can contain one or more containers that share storage, network, and a specification for how to run the containers.

182. What is a Node in Kubernetes?

A Node is a worker machine (physical or virtual) where Kubernetes runs Pods.

183. What is a ReplicaSet?

A ReplicaSet ensures that a specified number of identical Pods are running at all times. It automatically replaces failed Pods.

184. What is a Deployment?

A Deployment provides declarative updates to Pods and ReplicaSets. It manages rollout, rollback, and scaling of applications.

185. What is a Service in Kubernetes?

A Service is an abstraction that defines a logical set of Pods and a policy to access them.

Types:

- ClusterIP (default, internal)
- NodePort (external via node IP)
- LoadBalancer (cloud provider)
- ExternalName (DNS mapping)

186. How does Kubernetes perform load balancing?

Kubernetes Services distribute traffic among Pods using kube-proxy, which maintains iptables or IPVS rules.

187. What is etcd?

etcd is a distributed key-value store used by Kubernetes to store cluster state and configuration data.

188. What is a Namespace in Kubernetes?

Namespaces provide a way to divide cluster resources between multiple users or projects. Useful for isolation and organization.

189. What is Docker Volume?

A Docker volume is a persistent storage mechanism used to store data outside of a container's writable layer, making it independent of the container lifecycle. Volumes are managed by Docker and are stored on the host system in a special directory (/var/lib/docker/volumes/ by default).

190. Why Use Volumes?

- Persistence: Data remains intact even after container is removed.
- Sharing: Share data between multiple containers.
- Backup: Volumes can be easily backed up and restored.
- Performance: Better performance than bind mounts on most systems.

191. Basic Commands of Docker Volume?

Create a volume

```
docker volume create mydata
```

List all volumes

```
docker volume ls
```

Inspect a volume

```
docker volume inspect mydata
```

Remove a volume

```
docker volume rm mydata
```

Unused volumes can pile up. Clean them

```
docker volume prune
```

192. What is Docker Compose? Why is it used?

Docker Compose is a tool for defining and running multi-container Docker applications. It uses a YAML file (docker-compose.yml) to configure the services, networks, and volumes your application needs.

193. How do you start all services defined in a Compose file?

```
docker-compose up
```

194. How do you stop and remove all running services and networks created by Docker Compose?

```
docker-compose down
```

195. What is the difference between docker-compose up and docker-compose up --build?

- docker-compose up: Starts the services using existing images.
- docker-compose up --build: Rebuilds the images before starting containers.

196. How do you view logs for all services?

```
docker-compose logs
```

197. How do you link services together in Docker Compose?

By placing services under the same network (default or custom). They can communicate using the service name as the hostname.

198. What is the purpose of the depends_on keyword?

It defines the startup order of containers. For example:

```
yaml
```

```
depends_on:
```

```
- db
```

199. How can you pass environment variables to a container in Docker Compose?

```
environment:
```

```
- MYSQL_ROOT_PASSWORD=root
```

200. What is the difference between build and image in a Compose file?

- build: Tells Docker Compose to build an image from a Dockerfile.
- image: Specifies a prebuilt image (locally or from Docker Hub).

201. What is Terraform and why is it used in Azure?

Terraform is an Infrastructure as Code (IaC) tool used to automate provisioning and management of Azure resources in a consistent and repeatable manner.

202. What are Providers? How does AzureRM provider work?

The AzureRM provider is the plugin that allows Terraform to interact with Azure services.

203. What is Terraform State?

Terraform maintains a state file that keeps track of resources created in Azure.

204. Where should Terraform state be stored in Azure?

Azure Storage Account → Blob Container (terraform.tfstate file).

205. What is a Terraform backend?

Backend defines where Terraform state is stored (local, Azure blob, S3, etc.).

206. What is Terraform State Lock?

Prevents multiple users from running Terraform at the same time, avoiding conflicts.

207. How does Azure handle state lock?

Azure uses a Blob lease to lock Terraform state.

208. How to unlock a stuck state lock?

Use CLI to break blob lease:

```
az storage blob lease break --container-name tfstate --blob terraform.tfstate
```

209. What are .tf and .tfvars files?

- .tf: Terraform configuration files
- .tfvars: Variable definitions

210. Structure of common Terraform files

- main.tf → Resources
- variables.tf → Variable definitions
- outputs.tf → Outputs

211. What is terraform.tfstate and terraform.tfstate.backup?

State file and its auto-generated backup.

Common Commands:

terraform init

terraform plan

terraform apply

terraform destroy

terraform validate

terraform state list

terraform state rm

212. terraform init

Initializes backend and downloads providers.

213. terraform plan

Shows the execution plan.

214. terraform apply

Applies the changes.

215. How to run Terraform in Azure DevOps pipelines?

Using tasks for:

- Installing Terraform
- Azure CLI login
- Running init/plan/apply

216. How to secure backend credentials?

Store secrets in Azure Key Vault.

217. How to reference Azure resources in Terraform?

azurerm_resource_group.name

218. How to define dependencies?

- Implicit: via resource references
- Explicit: depends_on = []

219. What is a Terraform module?

Reusable infrastructure components.

220. How to call a module?

```
module "network" {  
  source = "../modules/vnet"  
}
```

221. What are Registry modules?

Modules available publicly on Terraform Registry.

222. What are provisioners?

Scripts used to configure VMs (not recommended).

223. How do you import existing Azure resources?

```
terraform import azurerm_resource_group.rg \  
/subscriptions/<id>/resourceGroups/myRG
```

224. Does Terraform generate .tf files after import?

No. You must manually write the .tf configuration.

225. Resource already exists

Fix → Use terraform import.

226. Provider version conflict

Fix → Declare provider version in required_providers.

227. Invalid index / null value

Occurs due to incorrect variable values.

228. How to authenticate Terraform to Azure?

- Azure CLI
- Service Principal

229. AzureRM provider configuration

```
provider "azurerm" {  
  features {}
```

}

230. What is the features {} block?

A mandatory block for AzureRM provider, even if empty.

231. What is Ingress in Kubernetes?

Answer:

Ingress is a Kubernetes API object that manages external HTTP/HTTPS access to services inside a cluster. It provides routing rules based on hostnames and paths.

232. What is an Ingress Controller?

Answer:

An Ingress Controller is a controller pod that watches Ingress resources and implements the routing rules using a reverse proxy like NGINX, Traefik, or Azure Application Gateway.

👉 *Ingress = rules*

👉 *Ingress Controller = engine that enforces those rules*

233. Can Ingress work without an Ingress Controller?

Answer:

✗ No.

Ingress does nothing by itself. Without an Ingress Controller, traffic will not be routed.

234. Difference between Service and Ingress?

Service

Ingress

Exposes pods internally/externally Routes HTTP/HTTPS traffic

Supports TCP/UDP

Supports Layer 7 (HTTP/HTTPS)

NodePort / LoadBalancer

Works with one LoadBalancer

235. Why do we need Ingress in AKS?

Answer:

Ingress helps:

- Reduce cost (single LoadBalancer)
- Centralize routing rules
- Enable SSL termination
- Provide path & host-based routing

236. Which Ingress Controllers are commonly used in AKS?

Answer:

- NGINX Ingress Controller (most common)
- Azure Application Gateway Ingress Controller (AGIC)
- Traefik
- Istio Ingress Gateway

237. How is SSL handled in Ingress?

Answer:

- SSL certificates stored as Kubernetes Secrets
- Ingress references the secret
- Controller performs SSL termination

238. Can we use one LoadBalancer for multiple apps?

Answer:

 Yes.

Ingress allows multiple applications to share one LoadBalancer using routing rules.

239. Explain path-based routing with Ingress

Answer:

Ingress can route traffic based on URL paths:

/path1 → app1-service

/path2 → app2-service

240. Explain host-based routing

Answer:

app1.example.com → app1-service

app2.example.com → app2-service

241. How does traffic flow in AKS using Ingress?

Answer:

1. Client sends request

2. Azure LoadBalancer receives traffic
3. Traffic forwarded to Ingress Controller pod
4. Ingress Controller routes to service
5. Service forwards to pod

242. Difference between Ingress and Azure Load Balancer?

Answer:

Azure Load Balancer Ingress

Layer 4

Layer 7

No routing logic

Path & host routing

One service

Multiple services

243. Can Ingress expose non-HTTP services?

Answer:

 No.

Ingress is HTTP/HTTPS only.

For TCP/UDP → use Service type LoadBalancer.

244. What happens if Ingress Controller pod goes down?

Answer:

- Traffic will fail
- Use replicas for HA
- Deploy controller as Deployment with multiple pods

245. Can we use multiple Ingress Controllers in one AKS cluster?

Answer:

 Yes, using IngressClass.

246. What is IngressClass?

Answer:

IngressClass defines which controller should handle the Ingress.

247. Your Ingress is created but the EXTERNAL-IP is pending. Why?

Answer:

Possible reasons:

- Ingress Controller Service is not LoadBalancer
- Azure quota limits exceeded
- Subnet has no free IPs
- AKS not allowed to create public IP

248. How do you expose multiple microservices using a single domain?

Answer:

Use path-based routing:

example.com/api → api-service

example.com/web → web-service

249. How do you scale Ingress Controller?

Answer:

- Increase replicas
- Enable HPA
- Ensure enough node capacity

250. How do you check which Ingress Controller is handling an Ingress?

Answer:

- Check ingressClassName
- Inspect controller logs
- Describe Ingress resource